



MSML610: Advanced Machine Learning

MSML610 Class Mechanics

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- ***MSML610***
- Class Map

Invariants of a Class Lecture

- **Invariants**

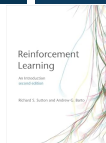
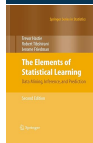
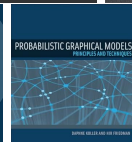
- Focus on intuition over math (unless necessary)
- Emphasize realistic assumptions and numerical methods
 - * Analytical solutions are so 1800s
- Interactive Jupyter notebook tutorials for hands-on approach
 - * Tutorials are mainly done at home
 - * Videos of each tutorial will be added over time

- **Class flow**

- Lessons alternate between slides, whiteboard, tutorials
- 2:45 hours per class lessons
 - * 50 mins
 - * 10 break
 - * 50 mins
 - * 10 mins
 - * 45 slides (Topic refresher!)

Books of the Class

- **Goal:** make the slides self-sufficient from recommended books
 - **Simple**
 - * Burkov: “*The Hundred-Page Machine Learning Book*” (2019)
 - * Burkov: “*Machine Learning Engineering*” (2020)
 - **Medium**
 - * Abu-Mostafa et al.: “*Learning From Data*” (2012)
 - * Martin: “*Bayesian Analysis with Python*” (2nd ed, 2021)
 - * Russell et al.: “*Artificial Intelligence: A Modern Approach*” (4th ed, 2020)
 - **Hardcore**
 - * Hastie et al.: “*The Elements of Statistical Learning*” (2nd ed, 2009)
 - * Koller et al.: “*Probabilistic Graphical Models: Principles and Techniques*” (2009)



Grading

- **Quizzes** (40%)
 - Multi-choice quizzes on previous 2 lessons
 - 4-5 quizzes to make you study during the semester and don't cram
- **Final Project** (60%)
 - A comprehensive application of course concepts
 - Python project selected from a list of topics

Class Projects

- The project is “*Build X with Y*”, where X is a “use case” and Y is a “technology”
 - Study and describe technology Y
 - Implement a use case X using the technology Y
 - Create Jupyter notebooks to demo your project
 - Commit code to GitHub and contribute to open-source repo
 - Write a blog entry
 - Present your project in a video
- There is a list of X and Y you can pick from, e.g.,
 - Statistical learning
 - Big data
 - LLMs
 - Deep learning
 - ...
- Each project:
 - Is individual or group ($n < 4$)
 - Has different levels of difficulty

Links

- [ELMS](#)
- [Syllabus](#)
 - Schedule
 - GitHub project
 - Class FAQs
- [Project specs](#)

Yours Truly

- **GP Saggese**
 - 2001-2006, PhD / Postdoc at the University of Illinois at Urbana-Champaign
 - [LinkedIn](#)
 - gsaggese@umd.edu
- **University of Maryland:**
 - 2023-, Lecturer for UMD DATA605: Big Data Systems
 - 2025-, Lecturer for UMD MSML610: Advanced Machine Learning
- **In the real-world**
 - Research scientist at NVIDIA, Synopsys, Teza, Engineers' Gate
 - 3x AI and fin-tech startup founder (ZeroSoft, June, Causify AI)
 - 20+ academic papers, 2 US



- MSML610
- *Class Map*

1. Intro

- A Map of Machine Learning
- What Is Artificial Intelligence
 - AI
 - Machine Learning
 - AI vs ML vs Deep-Learning
 - The Foundation of AI
 - Brief History of AI
 - AI State of the Art
 - Risks and Benefits of AI

2. Machine Learning Techniques

- Paradigms
- Techniques
 - Machine Learning in Practice
 - How to Do Research
 - * Simple Is Better
 - * Research Methodology
 - Pipeline Organization
 - Input Processing
 - Learning Algorithms
 - * Gradient Descent
 - * Stochastic Gradient Descent
 - Performance Metrics
 - * Precision and Recall
 - Model Selection
 - Aggregation
 - * Bagging
 - * Boosting
 - * Stacking

3. Knowledge Representation

- Knowledge Representation
 - Basics of Knowledge Representation
 - Examples of Logic
 - Logical Agents
 - Ontologies
 - Reasoning in Ontologies
- Propositional logic
- First-order Logic
- Non-classical Logics
- Description Logics
 - Semantic Web

4. Machine Learning Models

- Models
 - Naive Bayes
 - Decision trees
 - Random forests
 - Linear models
 - Perceptron
 - Logistic regression
 - LDA, QDA
 - Kernel methods
 - Support vector machines
 - Similarity-based models
 - Clustering
 - Anomaly detection

5. Machine Learning Theories

- Is machine learning possible?
- Growth function
- The VC dimension
- Overfitting
- Bias Variance Analysis
- Learning curves
- Learn-validation approach
 - Train / test
 - Cross-validation

6. Bayesian Statistics

- Logic-Based AI Under Uncertainty
- Probabilistic Reasoning
 - Conditional Independence
 - Bayesian Networks
 - Semantics of Bayesian Networks
 - Constructing a Bayesian Network
 - Exact Inference in Bayesian Networks
 - Approximate Inference in Bayesian Networks
 - * Direct sampling methods

7. Probabilistic Programming

- Concepts
- Coin Example
 - Analytical Approach
 - Communicating a Bayesian Analysis
 - Probabilistic Programming
- Posterior-Based Decisions
 - Chemical Shift: Example
 - Posterior Predictive Checks
 - * Robust Inference
- Groups Comparison
- Hierarchical Models
- Simple Linear Model
 - Logistic Regression
- Multiple linear regression
- Comparing Models
 - Posterior Predictive Checks
 - The Balance Between Simplicity and Accuracy
 - Measures of Predictive Accuracy
 - * Information Criteria
 - * Cross-Validation

8. Reasoning Over Time

- Reasoning over time
- HMMs
- Markov random fields
- Markov logic network
- State space models and Kalman filter
 - G-h filter
 - Discrete Bayes filter
- Dynamic Bayesian networks
- State space model
- Variational Inference
 - Expectation-Maximization (EM) Algorithm

9. Causal Inference

- Causal AI
 - Why Causal AI?
 - Concepts in Causal AI
 - Variables
 - Paths
 - The Ladder of Causation
 - Correlation vs causation models
- Business processes around data modeling
 - Modeling processes
 - Roles

10. Timeseries Forecasting

- Time Series
 - Basic definition
 - Time series operators
 - Time series decomposition
- Classical Methods
 - Simple models for stochastic process
 - Autoregressive models
 - Moving average models
 - ARMA(p , q) process
 - ARIMA model
 - ARCH model

11. Probabilistic Deep Learning

- Neural networks
 - Biological inspiration
 - Neural networks
- Advanced Neural Network Architectures
 - Convolutional networks
 - Recurrent Neural Networks (RNNs)
 - Deep learning learning algorithms
 - Deep learning architectures
- Fundamentals of Deep Learning
- Training Deep Neural Networks
- Interpretability and Explainability
- Deep Generative Models
- Bayesian Deep Learning
- Deep Probabilistic Models
- Uncertainty Quantification
- Probabilistic Programming and Inference
- Modern Research Frontiers
- Bonus Topics

12. Reinforcement Learning

- Sequential decision problems
 - Utilities over time
 - Algorithms for MDPs
- Reinforcement learning
 - Passive reinforcement learning
 - Active reinforcement learning
 - Generalization in reinforcement learning
 - Policy search
- Fundamentals
- Classical Methods
- Exploration Strategies
- Policy Gradient Methods
- Value Function Approximation
- Deep Reinforcement Learning
- Model-Based Reinforcement Learning
- Advanced Topics
- Applications

Refresher: Probability

- Probability
 - Probability definition
 - Probability measure
 - Independent events
 - Conditional probability
 - Law of total probability
 - Bayes theorem
- Random variables
 - Random variables
 - CDF, PMF, PDF of Random Variables
 - Joint distributions
 - Marginal distributions
 - Independent RVs
 - Conditional PDF RVs
- Mathematical expectation of RVs
 - Mean
 - Variance and covariance
 - Statistics of RVs
- Probability inequalities
- Statistical Inference

Refresher Probability Distributions

- Interesting RVs
 - Bernoulli
 - Binomial
 - Gaussian
 - Log-Normal
 - Poisson
 - Chi-square
 - Student's t-distribution
- Probability inequalities

Refresher Linear Algebra

- Linear algebra
 - Vector and vector spaces
 - Affine spaces
 - Vectors and matrices
 - Linear functions
 - Connections between Machine Learning and Linear Algebra

Refresher Information Theory

- Information theory
 - Entropy
 - * Kullback-Leibler divergence
 - Connections between Information Theory and ML

Refresher Game Theory

- Game theory
 - Connections between Machine Learning and Game Theory

Refresher: Numerical Optimization

- Optimization / numerical methods

Refresher: Stochastic Processes

- Stochastic processes